LESSON PLAN

NBCD (30)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	02
Total No. of Lesson Plans	6	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	By using antenna trainer, observe, measure output and radiation pattern on following antenna: - Half wave simple dipole antenna, Quarter wave
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	By using antenna trainer, observe, measure output and radiation pattern on following antenna: - Half wave simple dipole antenna, Quarter wave simple dipole antenna & Half wave folded dipole with reflector antenna.
5. Specific Objectives	30.1.1.1 Trainee should be able to practical on various lab FTL, RF & Embedded lab.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Operation of Antenna Trainer to measure output and observe radiation patterns for Half Wave Simple Dipole, Quarter Wave Simple Dipole, and Half Wave Folded Dipole with Reflector. Safety: Ensure RF power is off before connecting cables. Instructor supervision required during measurement.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the fundamental difference between a half-wave and a quarter-wave antenna?
(b) Define the term 'radiation pattern' in the context of antenna theory.
(c) List the three specific antenna types to be practiced in this session.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Setup of Antenna Trainer and Signal Generator • Connection of Half Wave Simple Dipole Antenna • Measurement of Output Voltage and Current • Observation of Radiation Pattern for Half Wave Dipole • Recording of Data for Half Wave Simple Dipole	L & Ds	PPT, IB	15 min
(b)	• Installation of Quarter Wave Simple Dipole Antenna • Measurement of Output Voltage and Current • Observation of Radiation Pattern for Quarter Wave Dipole • Comparison of Patterns between Half and Quarter Wave • Recording of Data for Quarter Wave Simple Dipole	L & Ds	PPT, PDF	20 min
(c)	• Assembly of Half Wave Folded Dipole with Reflector • Measurement of Output Voltage and Current • Observation of Radiation Pattern for Folded Dipole • Analysis of Reflector Effect on Pattern • Recording of Data for Folded Dipole with Reflector	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how the radiation pattern of a quarter-wave dipole differs from a half-wave dipole.
(b) Describe the role of the reflector in a folded dipole antenna system.
(c) Summarize the key measurements taken for each of the three antenna types.

12. Home Assignment and Follow Up	Prepare a brief report comparing the theoretical radiation patterns of the Half Wave Simple Dipole and the Half Wave Folded Dipole with Reflector based on the session observations.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Demonstration of real time data acquisition system.	Refer to the master syllabus reference list	LP 03

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min